

Cost-effectiveness of lenacapavir versus tenofovir disoproxil fumarate plus emtricitabine for pre-exposure prophylaxis to prevent HIV-1 transmission in gay, bisexual and other men who have sex with men in a European country

P. Wikman-Jorgensen^{1,2}, C. Iniesta^{3,4}, M. Ruiz-Alguero^{5,4}, F. Pulido⁶, J.A. Perez-Molina^{4,7}, J. Llenas-García^{2,4,8}
¹Hospital General Universitario de Elda-FISABIO, Infectious Diseases-Internal Medicine, Elda, Spain, ²Universidad Miguel Hernández, Clinical Medicine, Elche, Spain, ³Spanish Interdisciplinary AIDS Society (SEISIDA), Madrid, Spain, ⁴Instituto de Salud Carlos III, CIBER de Enfermedades Infecciosas (CIBERINFEC), Madrid, Spain, ⁵University of British Columbia, Faculty of Medicine, Vancouver, Canada, ⁶Institute for Healthcare Research Hospital 12 de Octubre (imas12), Universidad Complutense de Madrid, HIV Unit, Madrid, Spain, ⁷University Hospital Ramon y Cajal, Referral Centre for Tropical Diseases. Infectious Diseases Department., Madrid, Spain, ⁸Hospital Vega Baja-FISABIO, Infectious Diseases-Internal Medicine, Orihuela, Spain

Type selection
Preferred presentation type: Oral presentation

General data
Abstract category: Pre-exposure prophylaxis (PrEP)

Abstract body
Purpose: The study assessed which PrEP strategy was more cost-effective for gay, bisexual and other men who have sex with men (GBMSM) in Spain: daily oral TDF/FTC or subcutaneous LEN administered every six months.
Method: We fitted a dynamic Markov model that represents the dynamics of HIV-1 transmission in GBMSM in Spain(Fig 1). The model was calibrated to replicate epidemiological data from 2019 to 2023.

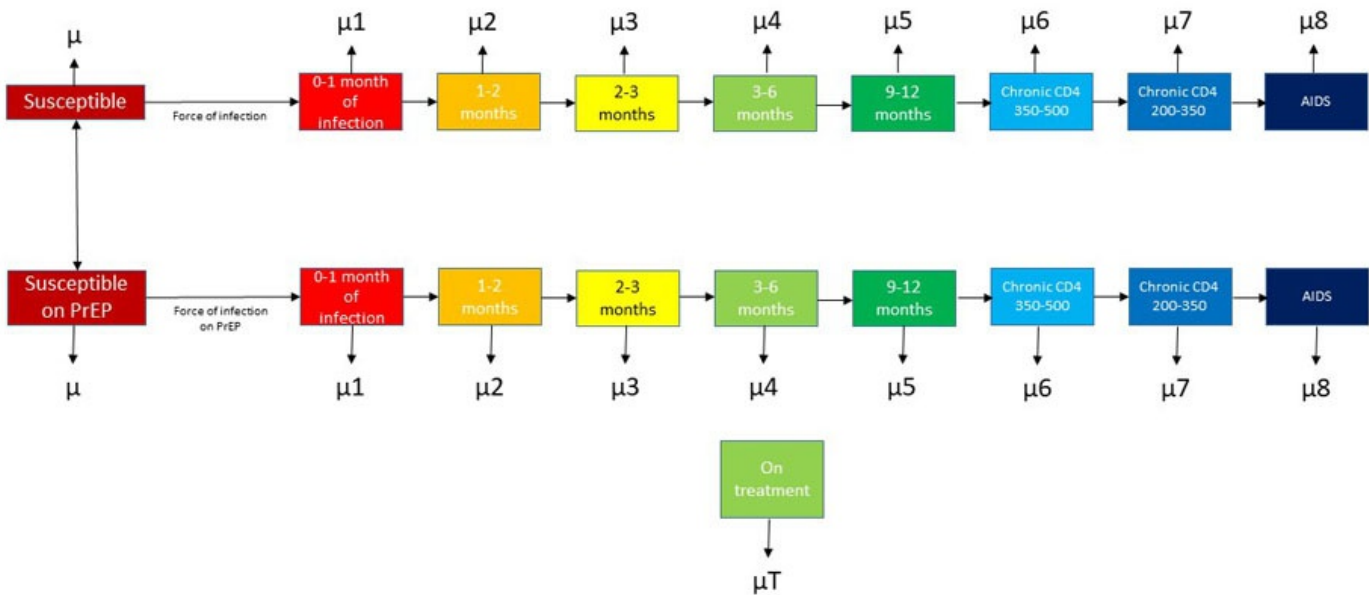


Fig 1. Compartmental model illustrating HIV disease progression with and without PrEP.

We used the perspective of the national health system and applied a 20-year time horizon. Quality-adjusted life years (QALY) were the health outcome variable, and the cost was accounted for in 2023 Euros(€). Our outcome variable was the incremental cost-effectiveness ratio (ICER). We used the 2023 gross domestic product per capita of Spain (€33,895) as the willingness-to-pay (WTP) threshold.

Key model Parameters	Value and OWSA range	PSA distribution
Number of GBMSM living with HIV	33558 (32000-38000)	Normal (33558,4000)
Baseline proportion of infected people on treatment	0.89(0.79-0.99)	Beta (0.96,0.1)
% GBMSM that go off PrEP	11.56% annually	NA
% GBMSM that start PrEP	30% annually	NA
Mortality of PLWHIV	>350	Normal (8.18e-06, 1.63e-06)
	200-350	Normal (0.000148, 2.95e-05)
	<200	Normal (6.41e-05, 1.28e-05)
Partner change rate in very high-risk compartment	52.5/year	Gamma (52.5/52, 52.5/52)

Partner change rate high-risk compartment	9/year	Gamma (2.5/52, 2.5/52)
PrEP effectiveness	TDx/FTC 86% risk reduction	NA
	LEN 96% risk reduction	

Results: The TDF/FTC arm yielded 8,546,030QALYs and cost €8,457,028,025 for the next 20 years. If LEN was used instead of TDF/FTC, it would cost €47,062,063,906 (increase of €38,605,035,881) producing 9,030,933QALYs (Increase of 484,903.7QALYs), yielding an ICER of 79,613.82€/QALYs. Adherence and number of GBMSM living with HIV were the main variables affecting the ICER in a deterministic sensitivity analysis. However, it did not come close to WTP. A probabilistic sensitivity analysis was also undertaken (Fig 2) where the cost was iterated down to €4,320 per six-month dose yielding a probability of cost-effectiveness of 93.6%.

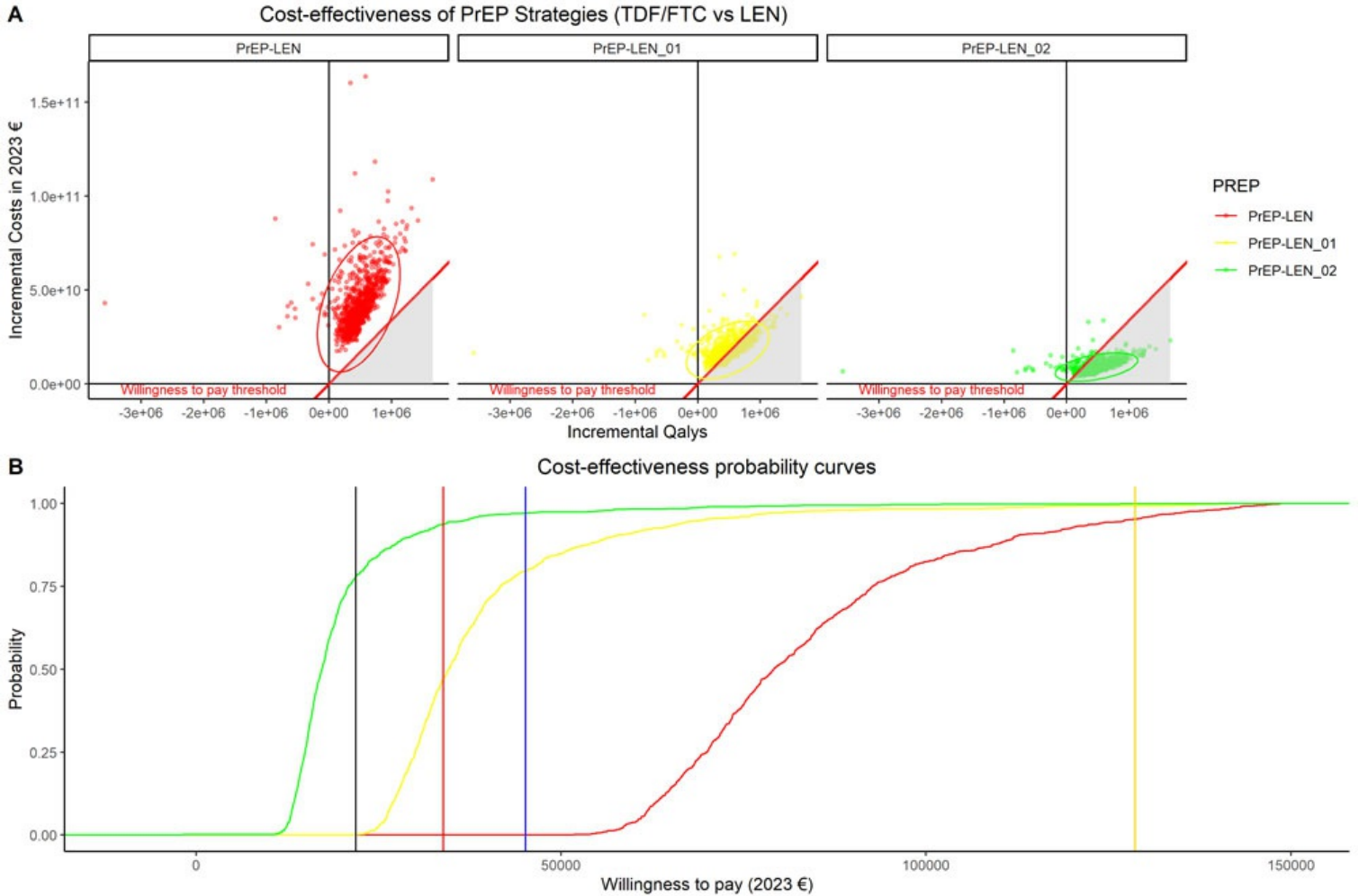


Fig 2. A) Cost-effectiveness planes. Each point represents a simulation, with 95% confidence ellipses. Red line indicates the willingness-to-pay threshold. B) Cost-effectiveness probability curves of the different drug costs evaluated. Sigmoidal curves: red(€21,088.5), yellow(€8,880), green(€4,320) Vertical lines; Black: GDP of Croatia; Red: GBP of Spain; Blue: GDP of Euro Zone, Yellow: GDP of Luxembourg.

Conclusions: LEN could be cost-effective in Spain at a price of €4,320 per six-month dose.

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